TITLE

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Abstract

Keywords: Multi-agent system, Vision-language model, Double reading, Error correlation, Visual question answering, Visual grounding

1. Introduction

Double reading of screening mammograms increased cancer detection compared with a single reader.[1] In diagnostic radiology, a systematic review found that a second reader changed reports at rates that depended on the setting.[2] Radiologists' errors are perceptual or interpretive: a reader misses an image finding or assigns an incorrect meaning to a finding.[3] A second reader adds information only to the extent that the readers do not make the same errors. Pooling independent diagnostic judgments improved accuracy when readers had similar individual accuracy.[4] The value of another reader therefore depends on both individual accuracy and error correlation. Artificial intelligence (AI) systems have also been evaluated in the second-reader position, including a simulation of double reading in breast cancer screening.[5]

Multi-agent systems built from large language models emulate a clinical consultation. Published medical systems assigned clinical roles to several instances of a model, which answered independently before discussing their answers or voting.[6,7] A reading panel applies this arrangement to questions about radiology images. A large evaluation of agent systems found that the benefit of collaboration shrank as the underlying model became more capable.[8] Another evaluation found correlated errors across language models, including models with different architectures and providers.[9] In a reading panel built from one vision-language model (VLM), every reader shares the model's weights. These include the visual encoder that extracts image findings and the language model that selects the answer. Sampling and role prompts change the text of each reader's response while leaving the model

and its image evidence shared. Different responses alone do not establish independent second reads.

VLMs answer many image questions using information in the question alone. The question-text prior is information supporting an answer from the question and options without image content. In general visual question answering, models exploited language priors; performance decreased when each question was paired with similar images requiring different answers.[10] This construction required the image to distinguish between answers to the same question. In medical visual question answering, leading multimodal models performed worse than random guessing on simple probing questions about diagnosis.[11] Analysis of medical image questions also found correct answers accompanied by flawed image comprehension.[12] Replacing the image cost medical VLMs at most 6.5 percentage points of accuracy on medical benchmarks, against 48 to 61 points on general-domain tasks,[13] and a radiology data set constructed to remove such shortcuts reduced text-only accuracy to near chance.[14] Answer accuracy alone does not measure the contribution of image evidence. In a reading panel, a prior that every reader shares is a common cause of the same error in all of them, and agreement among the readers does not show that each extracted the findings the answer requires.

Whether a reading panel of AI models provides independent second reads of radiology images has not been measured, nor have the sources of its shared errors. The aim of this study was to measure error correlation among AI readers answering radiology questions and determine its sources. We separated image evidence from the question-text prior and seeing from judging. Seeing was defined as the extraction of visual findings, and judging as the selection of an answer using findings and prior knowledge. We measured the contribution of each shared component: VLM, image rendering, role, reference cases, findings report, and discussion. The reader study evaluated four application programming interface (API) VLMs and five open-weight models on five item sets from VQA-RAD, SLAKE, and MedXpertQA.[15–17] Controlled manipulations varied the image through graded evidence, region occlusion, and paired images, and varied what readers shared, including the models used for seeing and judging (Fig. 1).

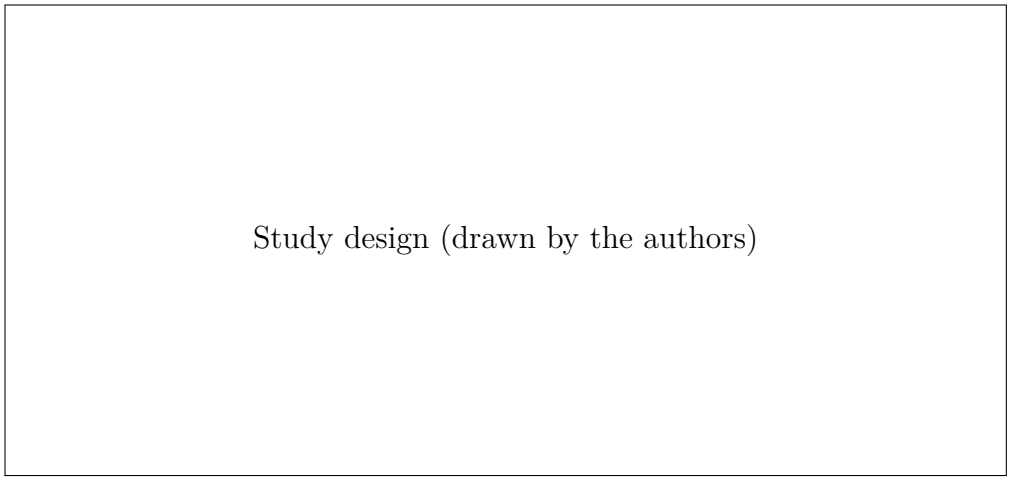


Figure 1: Study design. CAPTION

2. Material and methods

2.1. Study design

This reader study evaluated single readers and reading panels of AI VLMs answering closed questions about radiology images. Controlled manipulations varied the information available to readers and the components they shared. The design distinguished image evidence from the question-text prior and separated seeing from judging. The protocol and validation decisions were locked before the full runs. Pilot observations were excluded from the study analyses.

2.2. Data sets and sampling

Five item sets were assembled from public data sets. S1 contained 250 of 251 eligible VQA-RAD test questions with yes/no answers.[15] Duplicate image–question combinations were removed before sampling. S2 contained 250 of 355 eligible English SLAKE test questions labeled as closed and answered yes/no.[16] S3 contained 250 of 840 eligible MedXpertQA-MM items for which every attached image was a radiologic acquisition.[17] Image modality was labeled by gpt-4.1-nano, and claude-haiku-4.5 confirmed that candidate images were actual acquisitions. Drawings, diagrams, photographs, pathology slides, charts, and tables were excluded.

S4, SLAKE-target, contained 250 of 335 eligible English closed yes/no questions from the SLAKE training and test sets, excluding S2 items. Ques-

tions concerned an organ or lesion present in the image and localized by the supplied annotations. The expanded target bounding box occupied at most 40% of the image. Each item also had an equal-sized, nonoverlapping control box with at least 70% of its area inside the body. S5, SLAKE-pairs, contained 200 pairs, or 400 questions, spanning 104 question templates. Each pair used the same question with different images and opposite reference answers. Sampling allowed at most 5 pairs per template.

Sampling used stratified shuffling with seed 42. Strata were reference answer for S1, reference answer and modality for S2, body system and medical task for S3, and target type and modality for S4. S5 was stratified by question template. Yes/no option order was randomized separately for each item using its identifier. Images were resized proportionally, without upsampling, to at most 1024 pixels on the long side.

2.3. Models

The API models were gpt-5-mini, gemini-3.7-flash, claude-haiku-4.5, and qwen3.8-flash. All reader calls used OpenRouter; the routing and modality-screening outputs of gpt-4.1-nano (temperature 0) were generated once and reused. Where a model accepted the parameter, panel readers, discussion rounds, and report writers sampled at temperature 0.7 and the single reader at 0.3; the seed was 1, and gpt-5-mini used low reasoning effort. Google models were restricted to Gemini Flash. Routing and token budgets were held constant within each experimental condition. Because of its throughput, qwen3.8-flash ran only the original-image baseline (A0), graded evidence (A1), generic reading panel (C1), and varied image rendering (C2).

Five local open-weight models were evaluated: MedGemma-4B, Qwen3-VL-8B, Qwen2-VL-7B, HuatuoGPT-Vision-7B, and LLaVA-1.5-7B.[18–22] HuatuoGPT-Vision and LLaVA-1.5 shared the Contrastive Language–Image Pretraining (CLIP) ViT-L/14-336 vision encoder.[23] LLaVA-Med v1.5 was screened and excluded under the prespecified accuracy rule: its accuracy was 53%, below the 55% threshold, pooled over 100 validation items. These comprised 50 items each from S1 and S2. LLaVA-1.5 met the pooled threshold; its VQA-RAD validation accuracy was near chance.

2.4. Readers and reading panels

The single reader received a general-radiologist role and instructions to reason briefly, step by step, before selecting an answer. For each specialty reading panel, gpt-4.1-nano ranked 9 radiology subspecialty roles using the

question text. The roster covered neuroradiology, head and neck, thoracic, cardiovascular, abdominal, genitourinary, musculoskeletal, breast, pediatric, nuclear medicine, interventional, and emergency radiology. Panels of 1, 3, and 9 readers were nested: smaller panels used the first members of the same ranked roster and reused their initial answers. Answers were aggregated by majority vote.

Discussion panels started from these independent answers and allowed up to 3 subsequent rounds, stopping at unanimity. Readers reviewed peer answers and reasoning and could retain or revise their answers. Peer context was capped at 900 characters, with peer order shuffled by item and round. Generic panels contained 9 readers with identical general-radiologist prompts; their initial responses differed only through separate sampling.

2.5. Experiment A: image evidence and question-text prior

Experiment A used S1–S5. The original-image baseline compared the single reader, independent specialty panels, and discussion panels of 3 and 9 readers. Graded evidence compared original images capped at 1024 pixels, images capped at 256 or 128 pixels, and uniform gray images. Gray images retained the original rendered dimensions, and prompts and options remained unchanged. The single reader and independent specialty panels were evaluated at each evidence level.

On S4, the annotated target bounding box was expanded by 10% on each side and clipped to image boundaries. Target and control boxes were filled with mid-gray before resizing. Control boxes matched target-box dimensions, did not overlap the target, and met the body-coverage criterion. On S5, readers answered each member of the same-question, different-image pairs separately using the original images.

2.6. Experiment B: seeing and judging

Experiment B used S1–S3 and crossed 3 perceiver models with 3 reasoner models: gpt-5-mini, gemini-3.7-flash, and claude-haiku-4.5. Each perceiver received images without questions or answer options. It wrote findings reports of at most 120 words describing modality and plane, anatomy shown, and every abnormal or notable finding with its location. Reports were reused across questions about the same image. Each perceiver wrote its 9 reports for each image once, before any reasoner read a report. Every reasoner therefore read identical reports for a given perceiver, image, and slot. In total, 3 perceivers wrote 9 reports for each of 478 images (12,906 reports).

Reasoners received the report, question, and options without the image. In the independent-report condition, each of 9 readers received its own report; all perceiver–reasoner combinations were evaluated. In the shared-report condition, all readers received the first report; this condition used matching perceiver and reasoner models. Generic panels reading images directly provided the reference comparison. For S2, report coverage of the queried organ or lesion was assessed using a synonym list fixed from SLAKE labels before running this pathway.

2.7. Experiment C: shared components

Experiment C used S1–S3 to vary shared VLMs, image rendering, roles, reports, reference cases, and discussion records. In the rendering condition, generic readers 1–3 viewed original images, readers 4–6 viewed gamma-transformed images with gamma 0.7, and readers 7–9 viewed contrast-limited adaptive histogram equalization (CLAHE) images.[24] CLAHE used a clip limit of 2.0 and an 8×8 tile grid on the lightness channel. Gamma 0.7 replaced grayscale inversion because inversion reduced mean single-reader accuracy by 6.6 percentage points in validation, exceeding the prespecified 5-percentage-point limit. Cross-vendor comparisons combined existing reader responses without additional calls.

Reference-case memory was evaluated on S1 and S2 using VQA-RAD and English SLAKE training images with closed yes/no questions. Images identical to any test image were excluded by SHA-256 hash. For each query, exclusions also removed references within Hamming distance 8 of its 64-bit perceptual hash or with CLIP cosine similarity of at least 0.98. The data sets carried no patient identifiers; these bounds guarded against references from the same scan or patient.

Retrieval used CLIP ViT-L/14-336 image embeddings. The 27 nearest eligible images were shuffled deterministically by question identifier into 9 sets of 3. Shared memory supplied set 1 to every reader; private memory supplied set i to reader i. Each reference included its image and a closed question–answer pair. The question was selected by maximal term frequency–inverse document frequency (TF-IDF) cosine similarity to the current question. Text-only memory supplied the same question and answer without the reference image. Single readers also received image-based or text-only references.

2.8. Experiment D: open-weight models

Experiment D used S1–S5 and the same rendered pixels as the API arm. Conditions included original and gray images, 256- and 128-pixel images, gamma 0.7, CLAHE, and target and control occlusion on S4. S5 used original images only. Each item and condition required one forward pass without sampling. Option probabilities were obtained by applying softmax to next-token logits for the answer words yes/no on binary items and option letters otherwise. The highest logit among accepted capitalization and leading-space token variants represented each answer. Letter readout for yes/no items was rejected during validation because the LLaVA family assigned 96%–100% probability to "A" regardless of image content.

2.9. Outcome measures

Accuracy was the proportion of answers matching the reference. Error correlation, ϕ, was the mean Pearson correlation of binary error indicators across items over reader pairs. Pairs with constant error indicators were omitted. The effective number of independent readers was $N_{\text{eff}} = 9/(1+8\phi)$. For readers of unequal accuracy, correlation was also divided by its maximum given their marginal error rates, $\phi_{\max}$.

The accuracy difference between a reading panel and the single reader was divided into two parts that summed to it. The first was the mean accuracy of the panel's readers minus the single reader's accuracy. The second was the panel's majority-vote accuracy minus the mean accuracy of its own readers (aggregation). Both differences were paired by item. For every reading panel of 9 readers on yes/no items, observed majority-vote accuracy was compared with accuracy predicted by a beta-binomial model. This comparison covered VQA-RAD, SLAKE, SLAKE-target and SLAKE-pairs across all conditions. The model's mean was the readers' mean accuracy, and its intraclass correlation was their error correlation. A majority of 9 required at least 5 correct readers.

The prior decomposition cross-tabulated each reader's correctness with real and gray images. Consensus errors required at least 7 of 9 readers to be wrong; their majority answers were compared with gray-image majorities. Flip rate measured the proportion of answers changed by occlusion. Region dependence was target flip rate minus control flip rate, calculated for readers and panel majorities. Pair accuracy required both paired answers to be correct; the proportion of pairs answered differently was also measured. The two answers of each pair were compared as option text (yes or no) because

option order was randomized separately for each item. Report-pathway analyses compared accuracy and error correlation and separated errors by report coverage of the queried target. Reader and panel accuracies from reports were compared with those of the generic panel of the same reasoner model reading images directly. Differences were paired by item.

The primary analysis compared matched groups of reader pairs within each reasoner model using differences in mean pairwise error correlation. The role contrast was the mean for reader pairs in specialty panels minus that for reader pairs in generic panels. The image rendering contrast was the mean for pairs that saw different renderings minus that for generic-panel pairs. The reference cases contrast was the mean for pairs with private reference cases minus that for pairs with shared reference cases. The report contrast was the mean for pairs that each read their own report from the same model minus that for pairs reading one shared report. The perceiver model contrast was the mean for pairs reading reports from two different perceivers minus that for pairs reading different reports from one perceiver. The reasoner model contrast compared pairs of two reasoner models reading the same report with pairs of one reasoner model reading one shared report. The model contrast compared generic-panel readers of two models with generic-panel readers of one model. Both contrasts subtracted the mean for pairs of one model, averaged over the two models. The baseline was the error correlation between generic-panel readers of one model, which shared the model, image, rendering, role, and reference cases. Each contrast was computed for each model or model pair and averaged over models. An ordinary least squares regression of pairwise error correlation on indicators of the components two readers did not share summarized the same comparisons. The regression included one intercept per pair of reasoner models. Discussion was evaluated separately by comparing round-0 and final error correlation. Conformity was the proportion of readers correct in round 0 who ended the discussion wrong. It was stratified by whether their round-0 answer differed from the same reader's answer with the gray image. It was also stratified by whether the reader's round-0 answer was in the round-0 majority.

For open-weight models, visual reliance was the absolute change in log-odds of yes between real and gray images. For multiple-option items, reliance was the Kullback–Leibler divergence from the real-image to gray-image probability distribution. Region reliance subtracted the corresponding control-occlusion change from the target-occlusion change. Model-pair error correlations were compared by shared vision encoder, and Spearman correlation

related API consensus errors to open-weight visual reliance.

2.10. Statistical analysis

Percentile bootstrap 95% confidence intervals used 2,000 item-level resamples with seed 0, including paired differences. Error-correlation intervals used 500 resamples; the independence-budget regression, the matched contrasts, and the comparison of round-0 with final error correlation used 300 resamples. Matched contrasts were computed within the same resampled item sets as the regression. Resampling retained all reader responses and matched conditions for each item; paired-image measures resampled complete pairs. Unparsed answers counted as wrong. Abstentions (answers that could not be parsed, mostly refusals to read the gray image) were reported for each condition, and error correlation was also computed on the items both readers answered, as a sensitivity analysis. Analyses used descriptive estimates and confidence intervals without hypothesis tests or P values.

2.11. Validation before full runs

V1 checked API answer parsing and reasoning effort; V2 checked image transformations; V3 checked reference retrieval, leakage exclusions, and multiple-image parsing. V4 checked target and control boxes and image pairs; V5 checked report length, completeness, and answer parsing; V6 checked local-model probabilities and accuracy eligibility. Full runs used only the settings fixed by these checks.

2.12. Code and data availability

The code is public at `https://github.com/hq0709/consultation-arithmetic`. VQA-RAD, SLAKE, and MedXpertQA are public data sets.

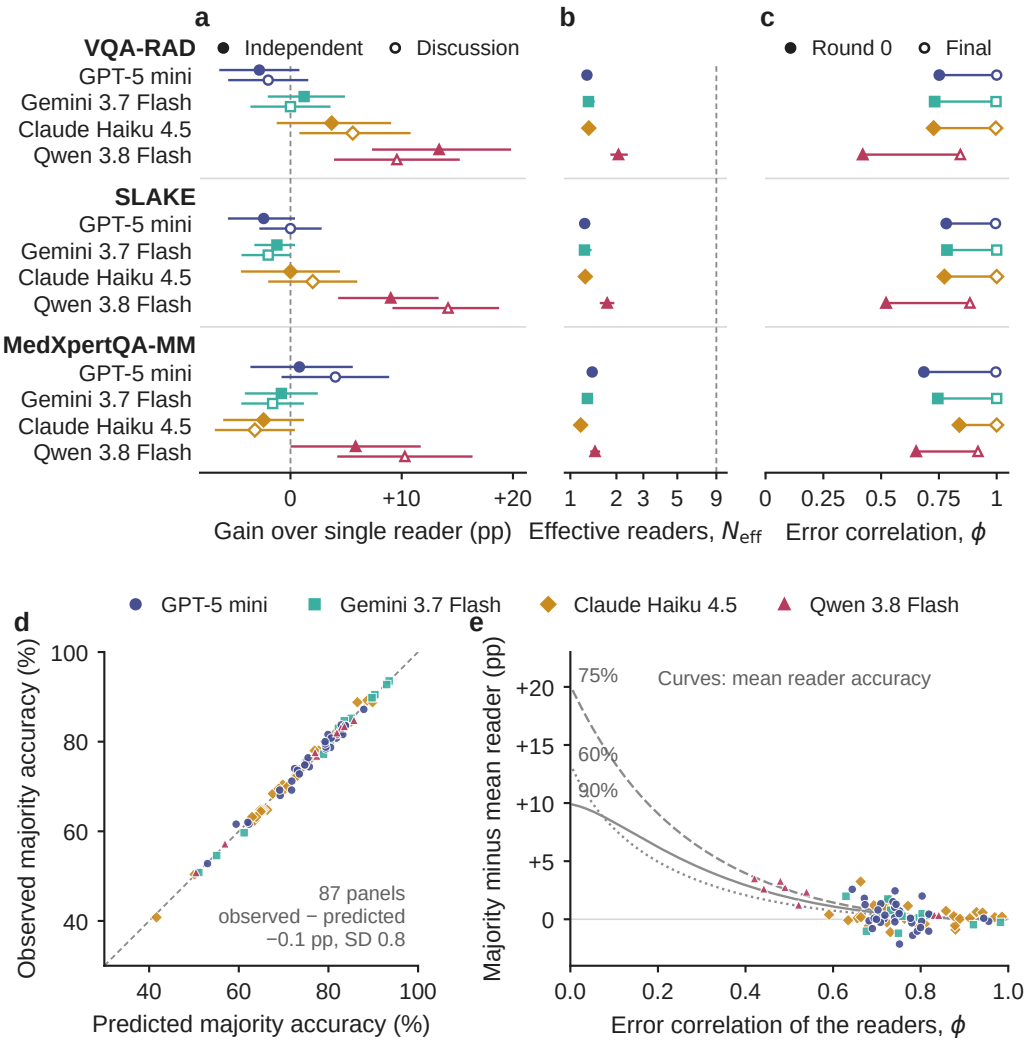


Figure 2: CAPTION

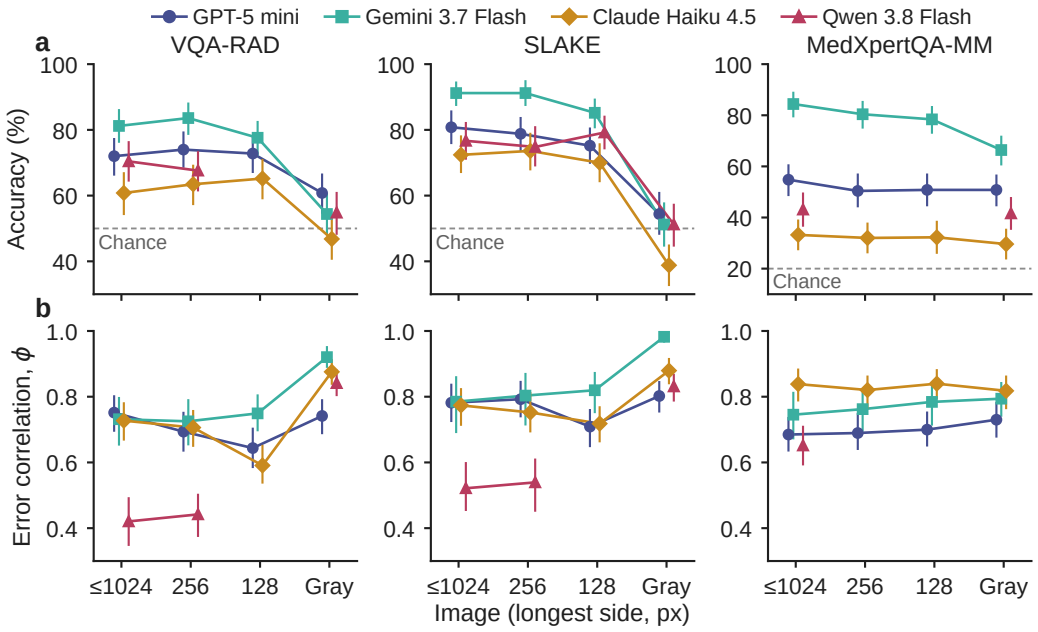


Figure 3: CAPTION

3. Results

3.1. Reading panels and error correlation

3.2. Image evidence and the question-text prior

3.3. Use of the queried region and of the image

3.4. Seeing and judging

3.5. What two readers share

3.6. Reference cases

3.7. Discussion among readers

3.8. Open-weight models

4. Discussion

A reading panel of nine readers built from one VLM did not provide nine independent reads. Across the original-image specialty panels, error correlation ranged from 0.685 to 0.836 for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5. The effective number of independent readers ranged from 1.17 to 1.39. Qwen 3.8 Flash had lower error correlation, 0.420–0.651, and

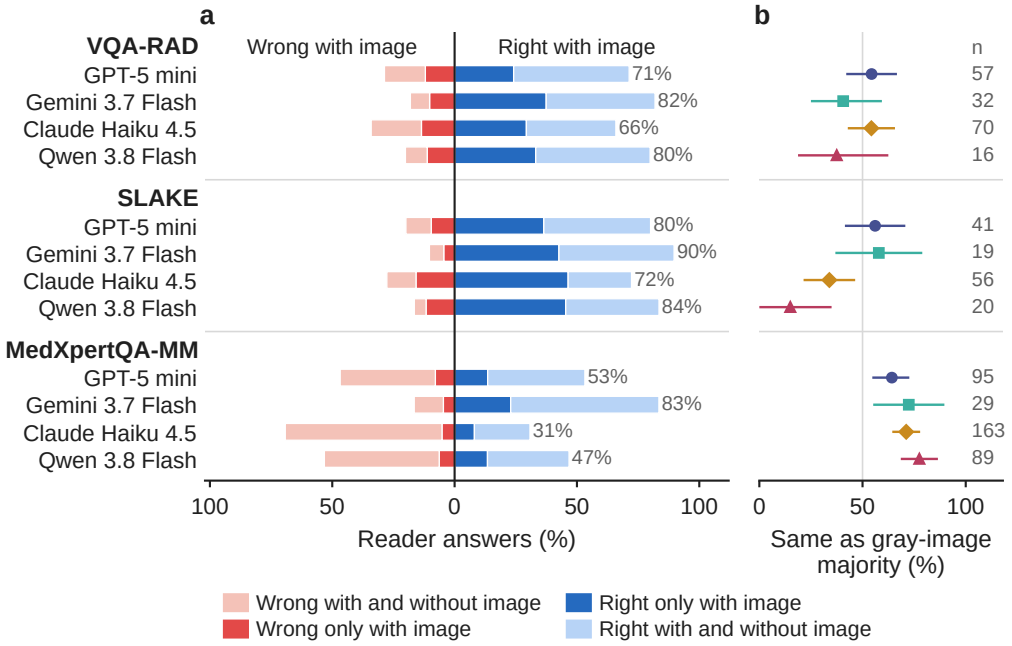


Figure 4: CAPTION

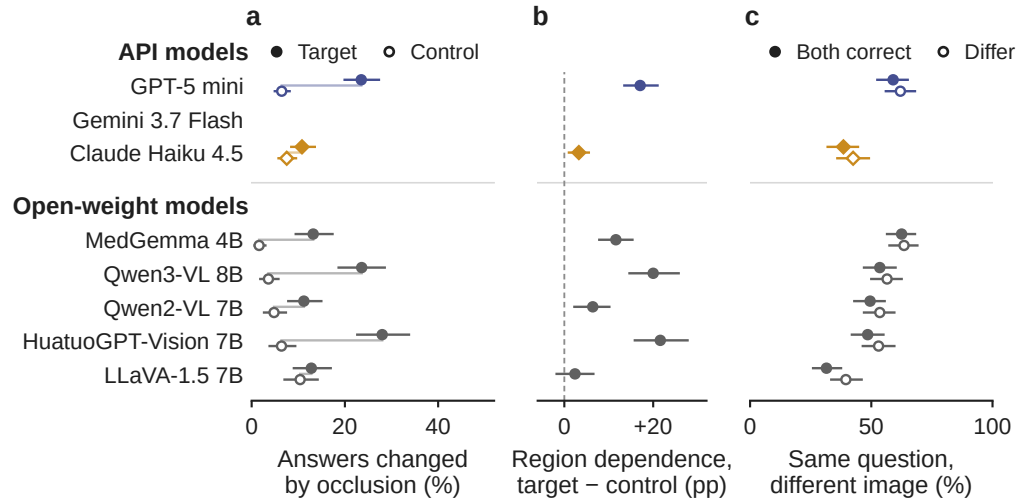


Figure 5: CAPTION

an effective number of independent readers of 1.45–2.06. The reading panels,

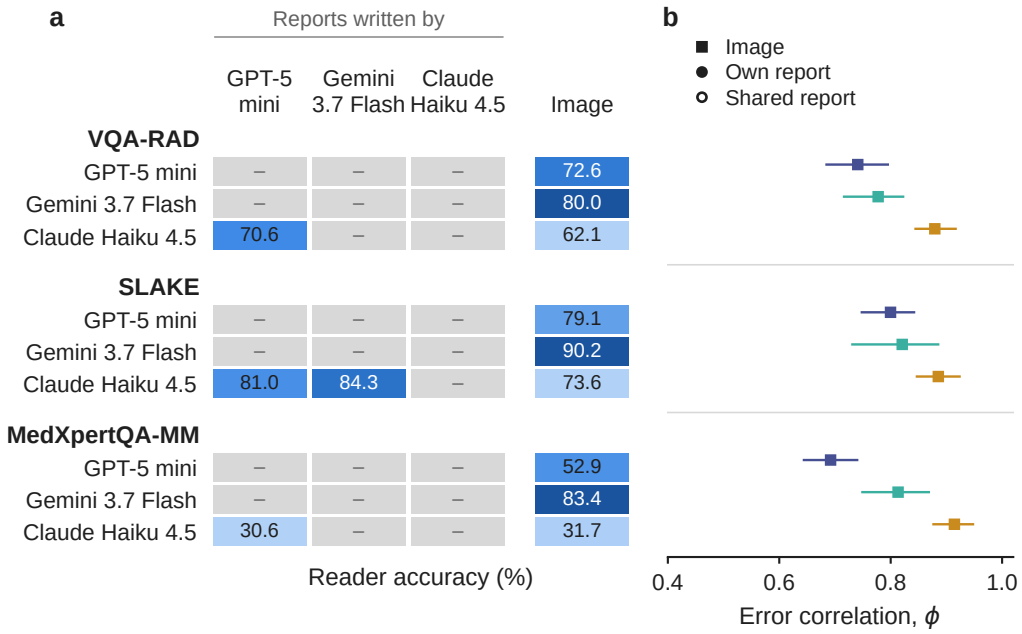


Figure 6: CAPTION

discussion, and panel arithmetic in Fig. 2 explain the limited return from adding readers. Across every nine-reader panel on yes/no items, observed majority accuracy closely matched the beta-binomial prediction from mean reader accuracy and error correlation. The observed-minus-predicted difference had a mean of -0.12 percentage points and an SD of 0.84 percentage points. Majority voting changed accuracy by -2.1 to +3.5 percentage points relative to the panel's own mean reader. Qwen's larger gain over the single reader came mainly from higher reader accuracy. On VQA-RAD, its mean panel reader exceeded the single reader by +9.7 percentage points (95% CI, +4.6 to +15.0). Aggregation added +3.5 percentage points (95% CI, +1.0 to +6.1). Reader accuracy and aggregation are distinct sources of improvement.

Discussion converged on agreement rather than consistently correcting errors. Final error correlation reached 0.843–1.000, and unanimity reached 85.2%–100.0%, yet 8.9%–70.0% of unanimous panels were wrong. The discussion and conformity analyses in Fig. 9 show how correct answers were lost during this convergence. Correct answers that differed from the gray-image answer were abandoned more often than correct answers that matched it, especially on MedXpertQA-MM. These image-dependent correct answers were

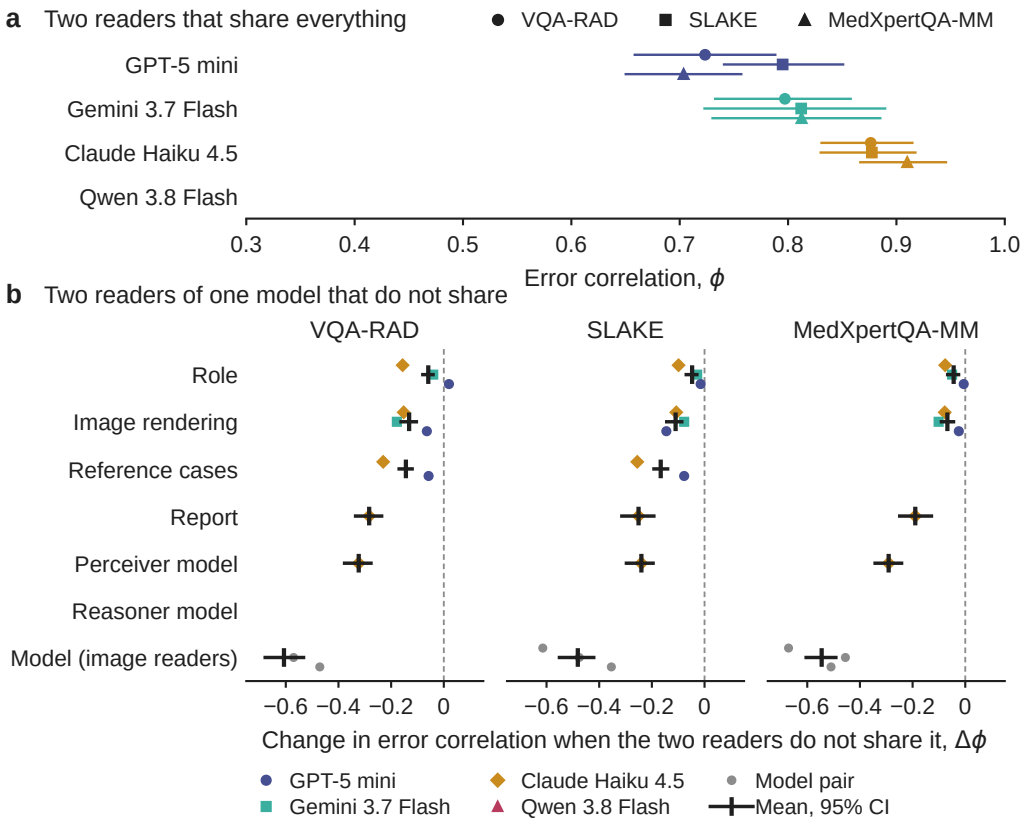


Figure 7: CAPTION

also more often held by a minority on that set. Discussion therefore exposed the answers most dependent on image evidence to pressure from a majority sharing the question-text prior. Minority correct answers were overturned in most model–data set combinations, whereas majority correct answers were usually retained. Agreement after discussion consequently reflects both initial evidence and deference to other readers. In mammography, radiologists followed incorrect AI BI-RADS suggestions, demonstrating deference even when the suggested assessment was wrong.[25] In the reading panels studied here, agreement among readers who share a prior carries that prior forward. Unanimity after discussion is therefore insufficient evidence of a correct reading.

The question-text prior was a common cause of shared errors. The graded-evidence experiment in Fig. 3 showed that removing the image low-

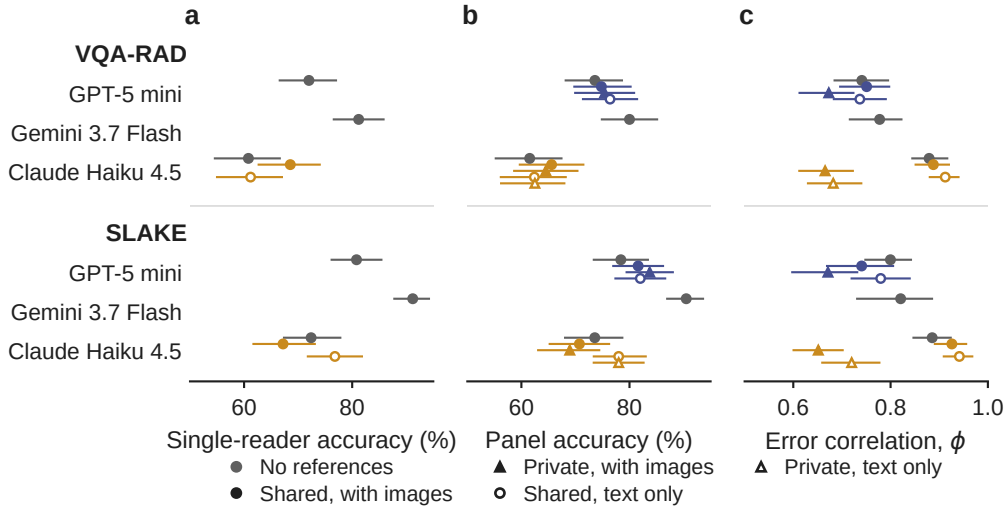


Figure 8: CAPTION

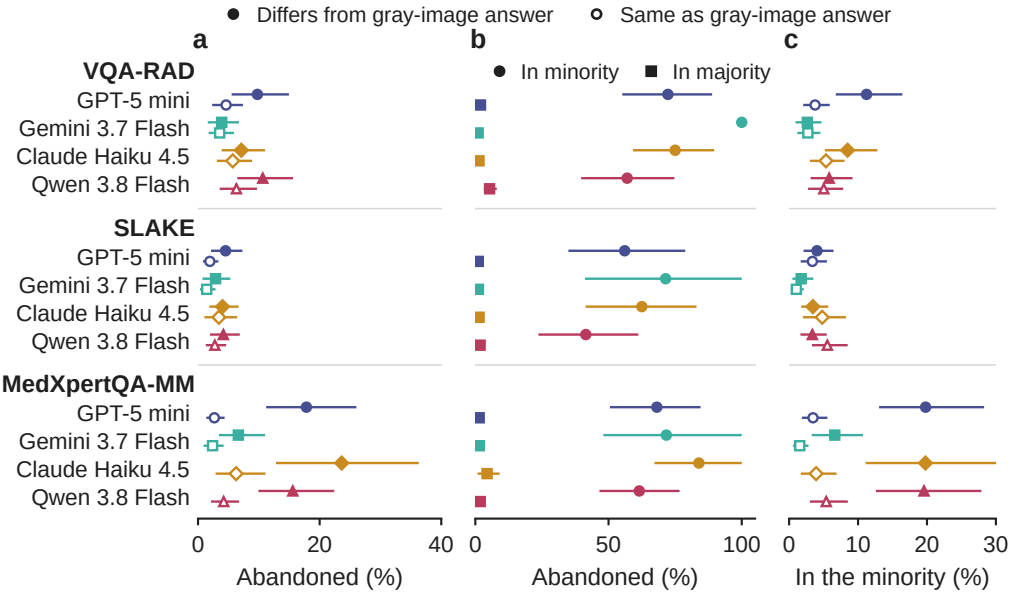


Figure 9: CAPTION

ered accuracy toward chance on VQA-RAD and SLAKE. On MedXpertQA-MM, it changed single-reader accuracy little for GPT-5 mini, Claude Haiku 4.5, and Qwen 3.8 Flash. Without the image, error correlation rose for Gem-

ini, Claude Haiku, and Qwen on VQA-RAD and SLAKE: readers converged on answers supplied by the remaining text. The reader-level decomposition in Fig. 4 connected these answers to errors with real images. Among GPT-5 mini, Gemini, and Claude Haiku, 33.9%–72.4% of consensus errors repeated the gray-image majority; Qwen's range was 15.0%–77.5%. Many reader answers were wrong both with and without the image, particularly on MedXpertQA-MM. Language priors in general visual question answering, poor performance under medical probing, and limited sensitivity to image substitution provide related evidence.[10,11,13] Clinical information also changes how diagnostic tests are read.[26] Here, the question text acts like clinical information that steers the reading. When it favors an incorrect answer, readers sharing the model also share a route to that error.

Readers used the image unevenly, even when they answered the same kinds of questions. Region occlusion in Fig. 5 changed GPT-5 mini answers substantially more for the queried region than for a matched control region. Its panel-reader region dependence was +17.1 percentage points (95% CI, +13.2 to +21.2), compared with +3.2 percentage points (95% CI, +0.8 to +5.7) for Claude Haiku. Open-weight models ranged from +2.4 to +21.6 percentage points. The image pairs in Fig. 5 tested whether the same question elicited opposite answers when the images required them. GPT-5 mini and Claude Haiku single readers gave different answers on only 42.5%–62.0% of pairs; their panel readers did so on 45.3%–66.0%. Open-weight models distinguished the paired images on 39.5%–63.5% of pairs. The remaining pairs received the same answer despite opposite reference answers. These interventions establish image use more directly than correctness on isolated questions. Accuracy alone overstates image use when a correct answer is supported by the text, consistent with reports of correct medical answers accompanied by flawed image comprehension.[12]

Seeing set the ceiling for judging from reports. In Fig. 6, question-blind reports written by Gemini 3.7 Flash raised accuracy for both complete reasoners, GPT-5 mini and Claude Haiku 4.5, on every set. Each reasoner's own reports did not consistently improve on reading the image directly. Separating seeing from the question therefore helped when the perceiver supplied better findings, rather than simply because an intermediate report was introduced. Readers sharing one report agreed almost completely on VQA-RAD and SLAKE, with error correlation of 0.944–0.986. Fixing the report removed most variation among readers of the same model, locating that variation primarily in seeing. On MedXpertQA-MM, shared-report error correlation was

lower, 0.800–0.893, and the reasoner also mattered. Given the same Gemini reports, GPT-5 mini reached 62.2% accuracy (95% CI, 57.2% to 67.2%), whereas Claude Haiku reached 45.6% (95% CI, 40.0% to 50.9%). Better findings improved both reasoners, but identical findings did not produce identical accuracy. The report pathway thus separates the contribution of extracted image findings from the reasoner's ability to select an answer using those findings.

Independent second reads require differences in what readers see or in the models that process the evidence. In Fig. 7, the mean contrasts over models lowered error correlation by small amounts for roles (-0.059 to -0.044), renderings (-0.128 to -0.069), and private reference cases (-0.166 to -0.146). A different perceiver lowered it more (-0.332 to -0.260), and a different model lowered it most (-0.583 to -0.487). These are changes in error correlation relative to the matched shared-component conditions. Lower error correlation alone did not guarantee higher panel accuracy. In Fig. 8, Claude Haiku's SLAKE panels with shared versus private text reference cases had error correlations of 0.941 (95% CI, 0.907 to 0.970) and 0.720 (95% CI, 0.657 to 0.779). Accuracy remained 78.0% in both conditions (95% CIs, 73.2% to 83.2% and 73.2% to 82.8%, respectively). For radiology practice, a panel of one model's readers should be counted as one read. An AI second read requires a different model or a different perceiver to supply additional independence. Human evidence likewise supports pooling independent judgments of similar accuracy,[4] and independent double reading improves mammographic cancer detection.[1] The relevant property is complementary errors at adequate reader accuracy.

The study measured answers to closed questions rather than clinical reporting. Items were yes/no or multiple-choice questions from public data sets. Each condition used one sampling seed, and the confidence intervals resampled items; they did not measure variability across sampling seeds. API models were evaluated at one time point through one router. Open-weight models had four to eight billion parameters. No human readers participated. Gray-image abstentions counted as wrong, so refusal to answer contributed to measured error. Error correlation on items answered by both readers was included as a sensitivity analysis.

Authors' contribution

TBD.

Ethics statement

The study used only public, de-identified data sets and involved no human participants. Institutional review board approval was not required.

Conflicts of interest

The authors declare no conflicts of interest.

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